PRESIDENCY UNIVERSITY

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Itgalpura, Rajankunte, Yelahanka, Bengaluru – 560064

MERN STACK HRMS PORTAL DEVELOPMENT

AN INTERNSHIP REPORT

Submitted by

BHARATHI NEERALAKERI – 20221ISE0104

Under the guidance of,

Ms. Pushpa C

BACHELOR OF TECHNOLOGY

IN

COMPUTER SCIENCE AND ENGINEERING (INFORMATION SCIENCE AND ENGINEERING)

GAIN MORE KNOWLEDGE
REACH GREATER HEIGHTS

PRESIDENCY UNIVERSITY

BENGALURU

APRIL 2026

PRESIDENCY SCHOOL OF COMPUTER SCIENCE AND ENGINEERING

BONAFIDE CERTIFICATE

Certified that this report **"MERN STACK HRMS PORTAL DEVELOPMENT"** is a bonafide work of **BHARATHI NEERALAKERI (20221ISE0104)**, who has successfully carried out the internship work and submitted the report for partial fulfilment of the requirements for the award of the degree of BACHELOR OF TECHNOLOGY in COMPUTER SCIENCE AND ENGINEERING (INFORMATION SCIENCE AND ENGINEERING) during 2025-2026.

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PRESIDENCY SCHOOL OF COMPUTER SCIENCE AND ENGINEERING

DECLARATION

I am a student of final year B.Tech in COMPUTER SCIENCE AND ENGINEERING (INFORMATION SCIENCE AND ENGINEERING), at Presidency University, Bengaluru, named BHARATHI NEERALAKERI, hereby declare that the internship work titled **"MERN STACK HRMS PORTAL DEVELOPMENT"** has been independently carried out by me and submitted in partial fulfillment for the award of the degree of B.Tech in COMPUTER SCIENCE ENGINEERING (INFORMATION SCIENCE AND ENGINEERING) during the academic year of 2025-2026. Further, the matter embodied in the internship has not been submitted previously by anybody for the award of any Degree or Diploma to any other institution.

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PLACE: BENGALURU

DATE: 22 - April 2026

INTERNSHIP OFFER LETTER

ADISYS
Observation Driven Insights

Date: 24/12/2025

To,
Bharathi Neeralakeri
Bengaluru, Karnataka

Subject: *Internship Offer – Software Engineer Intern*

Dear Bharathi,

We are pleased to offer you an Internship as a Software Engineer Intern at ADISYS Technologies Pvt. Ltd.

Based on your academic background and technical interest, you have been selected for a 4-month internship program, where you will gain practical exposure to software development and real-time project execution under the guidance of our technical team.

Internship Details:

- Position: Software Engineer Intern
- Duration: 4 Months
- Start Date: 10th January 2026
- End Date: 10th April 2026
- Work Location: ADISYS Technologies Pvt. Ltd., Bengaluru

INTERNSHIP STRUCTURE

- Exposure to software development life cycle (SDLC)
- Hands-on coding and development tasks
- Project-based learning and technical assignments
- Guidance and mentorship from senior engineers
- Periodic performance reviews

ADISYS Technologies Pvt. Ltd.
R & D Centre: # 256, 4th Main (above SBM) Mahalakshmi Layout, Bangalore - 560086 I Ph.: (+91) 080-23495922 / 9916912672
An ISO 9001:2008 Certified Company

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INTERNSHIP EXPECTATIONS

- Complete all assigned tasks and projects within timelines
- Maintain professionalism, discipline, and confidentiality
- Actively participate in learning and team discussions
- Adhere to company rules, policies, and code of conduct

COMPLETION BENEFITS

- Internship Completion Certificate
- Project Evaluation / Performance Feedback
- Potential consideration for future opportunities based on performance

This internship is offered strictly for **training and academic learning purposes** and does not guarantee permanent employment with ADISYS Technologies Pvt. Ltd.

We wish you a productive and enriching internship experience with our organization.

Warm regards,

For ADISYS Technologies Pvt. Ltd.

Raghunath Patil / Director Technical

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INTERNSHIP COMPLETION CERTIFICATE

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TO WHOMSOEVER IT MAY CONCERN

This is to certify that **Ms. Bharathi Neeralakeri** has successfully completed the internship at **ADISYS Technologies Pvt. Ltd.** from **10th January 2026** to **10th April 2026**. During the internship period, the candidate worked on the project **"Human Resource Management System (HRMS) using MERN Stack."**

During the internship, **Bharathi Neeralakeri** was sincere, hardworking, and completed all assigned tasks successfully. The candidate's performance was found to be satisfactory.

We wish all the success in her career

ADISYS TECHNOLOGIES PVT. LTD. * BANGALORE-86

For ADISYS Technologies Pvt. Ltd.

Date: 21st April 2026

Place: Bangalore

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We are grateful to **Dr. Md Ziaur Rahaman,** Internship Coordinator and **Mr. Sreehari T M,** Program Internship Coordinator, Presidency School of Computer Science and Engineering, or facilitating problem statements, coordinating reviews, monitoring progress, and providing their valuable support and guidance.

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BHARATHI NEERALAKERI

ABSTRACT

This report regarding my internship describes the knowledge, practical exposure, and technical skills I am gaining during my ongoing internship at Adisys Technologies Private Limited, a company focused on delivering innovative software solutions, web development services, and technology-based business applications.

During the initial phase of internship, I was introduced to modern web development technologies and full-stack application development concepts. I learned the fundamentals of frontend and backend development, database connectivity, user interface design, and responsive web applications.

As part of my internship training, I worked on the project titled **MERN Stack HRMS Portal Development**, where I gained hands-on experience in developing a Human Resource Management System using **MongoDB, Express.js, React.js, and Node.js**. This project helped me understand how full-stack applications are designed, developed, and deployed in real-time industry environments.

Currently, my learning is focused on modules such as employee management, attendance tracking, payroll processing, leave management, authentication systems, and dashboard development. I am also gaining knowledge about database management, REST API integration, role-based access control, and software testing practices used in organizations.

Overall, this internship is helping me strengthen my technical skills in MERN stack development, improve problem-solving abilities, and understand professional software development workflows, thereby preparing me for a successful career in the IT industry.

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ABBREVIATIONS

Abbreviation	Full Form
MERN	MongoDB, Express.js, React.js, Node.js
HRMS	Human Resource Management System
API	Application Programming Interface
REST	Representational State Transfer
JWT	JSON Web Token
NoSQL	Not Only Structured Query Language
UI	User Interface
UX	User Experience
CRUD	Create, Read, Update, Delete
HTTP	Hyper Text Transfer Protocol
JSON	JavaScript Object Notation
CSS	Cascading Style Sheets
HTML	HyperText Markup Language
RBAC	Role-Based Access Control
SDG	Sustainable Development Goals
SDLC	Software Development Life Cycle
MVC	Model View Controller
SPA	Single Page Application
npm	Node Package Manager
IDE	Integrated Development

Chapter 1

INTRODUCTION

The rapid growth of technology and digitization has transformed the way organizations manage their human resources. Traditional HR processes that were once handled manually — such as employee record keeping, attendance tracking, payroll management, and leave approvals — are now being replaced by automated, web-based solutions. A Human Resource Management System (HRMS) is a software platform that integrates all HR functions into a centralized digital application, improving efficiency, accuracy, and accessibility.

This internship project involves the design and development of a full-stack HRMS Portal using the MERN Stack — MongoDB, Express.js, React.js, and Node.js. The application was built to digitize and automate core HR operations for organizations, providing role-based access to Admin, HR Manager, and Employee users. This internship has provided hands-on experience in modern full-stack web development, RESTful API design, database management, and user interface design.

1.1 Background

Human Resource Management is a critical function in every organization, encompassing recruitment, employee management, attendance, payroll, and performance tracking. In many small and medium-sized organizations, these processes are still handled manually using spreadsheets or paper records, making them time-consuming and error-prone.

With the emergence of modern JavaScript-based full-stack frameworks, building robust and scalable web applications has become more accessible. The MERN stack — consisting of MongoDB (database), Express.js (backend framework), React.js (frontend library), and Node.js (runtime environment) — has become one of the most popular technology stacks for building modern web applications due to its flexibility, performance, and JavaScript-centric development approach.

This project was undertaken to build a comprehensive HRMS portal that handles employee registration, attendance, leave management, payroll, and departmental organization. The system ensures secure access through JWT-based authentication and role-based control, making it suitable for real-world enterprise deployment.

1.2 Statistics of Internship

The internship was focused on the end-to-end development of the MERN Stack HRMS Portal project. The following table summarizes the key statistics related to the internship:

Parameter	**Details**
Project Title	MERN Stack HRMS Portal Development
Technology Stack	MongoDB, Express.js, React.js, Node.js
Duration	3 Months
Organization / Platform	Adisys Technologies Pvt.Ltd
Role	Full Stack Developer Intern
Modules Developed	5+ (Employee, Attendance, Leave, Payroll, Department)
Authentication	JWT-based Role Authentication
Database	MongoDB (Atlas / Local)
Tools Used	VS Code, Git, GitHub, MongoDB Compass

Table 1.2: Key Statistics of Internship

1.3 Prior Existing Technologies

In the software development domain, organizations use various technologies, frameworks, and tools to build efficient, scalable, and user-friendly web applications. Before the implementation of modern full-stack technologies, many organizations relied on traditional systems such as desktop applications, standalone databases, spreadsheets, and manual record-keeping methods to manage employee information and HR operations.

Earlier Human Resource Management Systems (HRMS) were mainly developed using basic programming languages and monolithic architectures, where frontend, backend, and database functionalities were tightly coupled. These systems were difficult to maintain, lacked scalability, and provided limited user experience. Many small and medium organizations also depended on manual processes for attendance management, payroll calculation, leave tracking, and employee record maintenance.

With the growth of web technologies, organizations started adopting modern development frameworks and databases to improve performance and flexibility. Frontend technologies such as HTML, CSS, JavaScript, Bootstrap, and Angular were widely used for creating interactive user interfaces. Backend technologies such as PHP, Java, Python, and .NET were used for business logic and server-side operations. Databases such as MySQL, Oracle, and SQL Server were commonly used to store organizational data.

In recent years, the MERN Stack has become one of the most popular technologies for developing modern web applications. MERN consists of MongoDB, Express.js, React.js, and Node.js. MongoDB is used for flexible NoSQL database storage, Express.js simplifies backend API development, React.js enables dynamic and responsive user interfaces, and Node.js provides fast and scalable server-side execution.

In addition, organizations use development tools such as Visual Studio Code for coding, Postman for API testing, GitHub for version control, and MongoDB Compass for database management. These technologies help organizations build secure, efficient, and scalable systems while improving overall productivity and user experience.

1.4 Proposed Approach

The internship started with an orientation session at **Adisys Technologies Private Limited**, where interns were introduced to the organization, work culture, internship structure, and project objectives. During the initial phase, guidance was provided regarding software development practices, project workflow, and the technologies that would be used during the internship.

After the orientation, a training program was conducted for interns for the first few weeks, where the fundamentals of web development and full-stack technologies were explained. This training included basic concepts of **HTML, CSS, JavaScript, React.js, Node.js, Express.js, and MongoDB**. Interns were also introduced to coding standards, responsive design principles, and database management concepts.

Following the training phase, interns were assigned practical projects to apply their knowledge in real-time development environments. As part of my internship, I was assigned the project titled **MERN Stack HRMS Portal Development**. During this phase, I worked on developing different modules such as employee management, attendance tracking, payroll management, leave management, authentication system, and dashboard interfaces.

The internship then progressed into the implementation phase, where I worked under the guidance of mentors and senior developers. This phase involved frontend development using React.js, backend API creation using Node.js and Express.js, database integration using MongoDB, and testing the application modules.

Currently, I continue to enhance the project by improving user interface design, optimizing performance, fixing bugs, and learning advanced concepts in full-stack development. At the end of the internship, the completed project and overall performance evaluation will contribute to future career opportunities and professional growth in the software development field.

1.5 Objectives

The main objectives of this internship project are:

- To design and develop a full-stack HRMS web application using the MERN stack (MongoDB, Express.js, React.js, Node.js).
- To implement role-based access control (RBAC) with three user roles: Admin, HR Manager, and Employee.
- To develop core HR modules including Employee Management, Attendance Tracking, Leave Management, Payroll Processing, and Department Management.
- To build a secure authentication system using JSON Web Tokens (JWT) for session management and user verification.
- To design a responsive and user-friendly frontend interface using React.js and modern CSS frameworks.
- To develop RESTful APIs using Express.js and Node.js for seamless communication between the frontend and backend.
- To use MongoDB as the database for flexible, scalable, and efficient data storage.
- To gain practical experience in full-stack development, project management, and professional software development practices.

1.6 Sustainable Development Goals (SDGs)

The Sustainable Development Goals (SDGs) introduced by the United Nations serve as a global framework for achieving a better and more sustainable future. The MERN Stack HRMS Portal Development project aligns with several SDGs:

Figure 1.6 SDG

- SDG 4 – Quality Education: The internship provides practical training in full-stack web development, enhancing the technical skills of students beyond academic learning.

- SDG 8 – Decent Work and Economic Growth: Building an HRMS portal directly supports HR operations in organizations, promoting efficient workforce management and organizational productivity.

- SDG 9 – Industry, Innovation and Infrastructure: The project demonstrates innovation in software development by building a modern, scalable web application using the latest MERN stack technologies.

- SDG 17 – Partnerships for the Goals: The internship fosters collaboration between academic institutions and industry, bridging the gap between theoretical education and real-world application.

1.7 Overview of Internship Report

This internship report is organized into different chapters to describe the internship experience and the knowledge gained during the program. The first chapter provides an introduction to the internship, including the background of the organization, objectives, and overall context.

The following chapters explain the technologies and tools involved in the internship, such as the **MERN Stack** including MongoDB, Express.js, React.js, and Node.js, along with the development processes followed during the training period.

Later chapters describe the work carried out during the internship, including project development, module implementation, coding practices, testing, and practical exposure under mentor guidance. The report also discusses the challenges faced, the learning outcomes achieved, and finally concludes with the overall experience and future scope in the software development domain.

Chapter 2
LITERATURE REVIEW

The literature review provides a comprehensive understanding of existing research and technological advancements related to Human Resource Management Systems (HRMS) developed using modern web technologies. It helps in identifying previously implemented methods, their strengths and limitations, and the research gaps addressed by the proposed system. This chapter focuses on reviewing current HRMS systems, technological frameworks, software architectures, automation techniques, user-centered evaluation methods, and security and data management considerations in Human Resource Management.

2.1 Review of Existing Systems and Frameworks

Human Resource Management Systems are software applications designed to manage employee-related activities within an organization. In earlier days, HR operations were maintained manually using paper records, spreadsheets, and offline systems. These traditional methods were time-consuming, error-prone, and difficult to manage when the number of employees increased. As organizations adopted digital transformation, web-based HRMS solutions became more popular.

Modern HRMS platforms such as BambooHR, Zoho People, Workday, and SAP SuccessFactors provide integrated modules for employee management, attendance tracking, leave management, payroll processing, and performance monitoring. These systems improve efficiency by reducing manual tasks and providing centralized access to employee data. They also support report generation, notifications, and remote accessibility.

However, commercial HRMS software often requires high subscription costs and may contain unnecessary features for small and medium-sized businesses. Some organizations also require customized workflows that cannot be easily implemented in ready-made systems. Therefore, there is a need for cost-effective and customizable HRMS portals developed using modern technologies such as the MERN stack.

2.2 Technological Approaches and Architectures

Modern HRMS portals are generally developed using full-stack web technologies that include frontend, backend, and database layers. The MERN stack is one of the most effective combinations for developing such applications because it uses JavaScript throughout the system.

MongoDB is a NoSQL database used to store employee details, attendance logs, leave records, payroll information, and login credentials. It offers flexibility in data storage and supports scalable applications. Express.js is a backend framework that helps in creating REST APIs, handling routes, and managing server-side operations.

React.js is used to build interactive and responsive user interfaces. It allows developers to create reusable components such as dashboards, forms, tables, and navigation menus. Node.js provides the runtime environment for backend execution and supports asynchronous processing, making the application faster and more scalable.

The MERN architecture enables smooth communication between frontend and backend through APIs. It also simplifies maintenance and future feature expansion.

2.3 Algorithmic Approaches for Automation

Automation is an important feature in Human Resource Management Systems because it reduces repetitive manual work. HRMS portals implement rule-based logic and backend processing to automate daily HR operations.

Attendance management systems automatically calculate working hours based on check-in and check-out times. They can identify late arrivals, absences, and overtime. Monthly attendance summaries are generated without manual calculation.

Leave management modules automate the process of leave requests and approvals. Employees submit leave applications online, and managers can approve or reject them through the portal. The leave balance is updated automatically after approval.

Payroll systems use predefined formulas to calculate salaries. Basic salary, allowances, deductions, bonuses, and tax values are processed automatically to generate monthly payslips. This reduces accounting errors and saves time.

Role-based login systems are also implemented to control access. Administrators manage the entire portal, HR managers handle employee-related functions, and employees can view only their own data. These automation methods improve productivity and operational efficiency.

2.4 User-Centric Design and Evaluation

A successful HRMS portal must be user-friendly and easy to operate. Different users such as administrators, HR managers, and employees interact with the system daily, so the interface should be simple and responsive.

Administrators require dashboards showing employee count, department details, pending leaves, and payroll summaries. HR managers need pages for adding employees, marking attendance, approving leave requests, and managing reports. Employees need self-service features such as profile viewing, attendance status, leave applications, and salary slips.

Modern UI design principles focus on clean layouts, readable text, proper alignment, responsive pages, and easy navigation. React.js helps create dynamic dashboards and reusable components that enhance user experience.

System evaluation is based on ease of use, speed, accessibility, design quality, and user satisfaction. A well-designed HRMS portal increases productivity and reduces confusion among users.

2.5 Security and Data Management

HRMS portals store sensitive employee information such as salaries, addresses, attendance records, and login credentials. Therefore, proper security mechanisms are essential. Authentication systems verify users during login using email and password. Passwords should be encrypted using hashing algorithms such as bcrypt.

JWT authentication is commonly used in MERN applications to manage secure sessions. Authorization ensures that users access only permitted modules according to their role. For example, employees cannot modify payroll records, while administrators have full control.

Input validation protects the system from malicious attacks. Database backup and recovery mechanisms help prevent data loss. MongoDB provides efficient storage, indexing, and scalability features for secure data management.

2.6 Conceptual Framework

The proposed MERN Stack HRMS Portal follows a three-layer architecture. The frontend layer is developed using React.js and provides login pages, dashboards, forms, and employee interfaces. The backend layer is developed using Node.js and Express.js, which process requests, implement business logic, and provide secure APIs. The database layer uses MongoDB to store employee and organizational data.

Users interact with the frontend interface. Requests are sent to the backend server, processed securely, and stored or retrieved from MongoDB. The results are then displayed to the users in real time. This architecture ensures flexibility, maintainability, and future scalability.

2.7 Summary and Identified Gaps

The literature review shows that existing HRMS systems are useful but often expensive, complex, or not fully customizable. Small organizations need affordable systems that include only required features. Some older systems also lack modern dashboard designs and responsive user interfaces.

The proposed **MERN Stack HRMS Portal Development** aims to overcome these limitations by providing a secure, customizable, and scalable solution. It includes employee management, attendance tracking, leave handling, payroll processing, and role-based login features. The use of the MERN stack makes the system modern, efficient, and suitable for future enhancements.

Chapter 3
METHODOLOGY

This chapter explains the methodology and development approach followed during the MERN Stack HRMS Portal project. The project was built using a structured and iterative development process, ensuring that each module was properly designed, developed, tested, and integrated before moving to the next phase.

3.1 Overall Training Framework

The HRMS portal was developed using an Agile development methodology, where the project was divided into small sprints. Each sprint focused on a specific module or feature, allowing for incremental development and continuous testing. The overall development was organized into six major phases: requirement analysis, system design, backend development, frontend development, integration & testing, and deployment.

The development was carried out using Visual Studio Code as the primary IDE. Git and GitHub were used for version control and code management. Node.js and npm were used to manage dependencies and packages across both the frontend and backend.

3.2 SDLC Approach

The Software Development Life Cycle (SDLC) approach followed for this project was Agile with iterative sprints. The key phases of the SDLC adopted are described below:

- Requirement Gathering: Identifying HR modules, user roles, and system features through analysis of common HRMS workflows.
- System Design: Creating the database schema using MongoDB collections, designing RESTful API endpoints, and creating wireframes for the UI.
- Implementation: Coding the backend using Node.js and Express.js, and the frontend using React.js.
- Testing: Unit testing individual API endpoints using Postman, and integration testing of the full stack application.
- Deployment: Hosting the application and ensuring it is accessible to users.
- Maintenance: Iterative bug fixes and improvements based on testing feedback.

3.3 Tools and Technologies Used

The following tools and technologies were used during the development of the MERN Stack HRMS Portal:

Technology / Tool	Category	Purpose
MongoDB	Database	NoSQL database for storing HR data
Express.js	Backend Framework	RESTful API development
React.js	Frontend Library	Building UI components and SPA
Node.js	Runtime Environment	Server-side JavaScript execution
JWT	Authentication	Secure token-based user authentication
Mongoose	ODM Library	MongoDB object modeling for Node.js
Axios	HTTP Client	API calls from React frontend
Git & GitHub	Version Control	Code management and collaboration
VS Code	IDE	Code editor for development
MongoDB Compass	DB Management	GUI for managing MongoDB data
Tailwind CSS / Bootstrap	CSS Framework	Responsive UI design

Table 3.3: Development Tools and Technologies

3.4 Professional Learning Environment

The internship also provides an opportunity to work in a professional environment under the guidance of mentors and experienced team members. Interns are encouraged to learn continuously, improve coding skills, ask questions, and understand real-time software development processes followed in the organization.

Working on the MERN Stack HRMS Portal Development project helped me understand different areas of web development such as frontend design, backend development, database management, API integration, and testing. This professional environment enhanced my technical knowledge, teamwork, communication skills, and overall understanding of industry-level project development.

3.5 Summary

This chapter described the development methodology followed for the MERN Stack HRMS Portal project. An Agile approach with iterative sprints was used, combined with a three-tier MERN architecture. The tools and technologies used were selected based on their suitability for building a scalable, secure, and maintainable full-stack web application.

Chapter 4
PROJECT MANAGEMENT

Project management is an essential process in software development that ensures the successful planning, execution, monitoring, and completion of a project within the given schedule and budget. For the **MERN Stack HRMS Portal Development**, project management was followed to organize activities such as requirement gathering, system design, coding, testing, deployment, and documentation. Proper project management helps in improving productivity, reducing delays, controlling risks, and ensuring the delivery of a high-quality system. This chapter explains the project timeline, risk analysis, and project budget in detail.

4.1 Project Timeline

The project timeline represents the schedule followed to complete the development of the MERN Stack HRMS Portal. A clear timeline helps in dividing the project into manageable phases and ensures timely completion.

Timeline Explanation

During the first phase, the project requirements such as employee management, attendance, leave management, payroll, and role-based login were finalized. In the design phase, MongoDB collections and user interface layouts were planned.

The development phase included React.js frontend implementation and Node.js backend API integration. Testing was carried out after coding completion to ensure system reliability. Finally, deployment and project report preparation were completed.

This planned timeline ensured that the project was developed systematically within eight weeks.

Phase No.	Project Phase	Duration	Activities
1	Requirement Analysis	Week 1	Collecting requirements, identifying modules, defining user roles
2	System Design	Week 2	Database design, UI wireframes, architecture planning
3	Frontend Development	Week 3 - Week 5	Creating login page, dashboards, forms, navigation
4	Backend Development	Week 4 - Week 6	API creation, authentication, database integration
5	Testing & Debugging	Week 7	Functional testing, fixing bugs, UI improvements
6	Deployment & Documentation	Week 8	Final deployment, report preparation, screenshots

Table4.1: Project Development Schedule

4.2 Risk Analysis

Risk analysis helps in identifying possible issues that may affect the successful completion of the project. By understanding risks early, preventive actions can be taken. The following risks were considered during the development of the MERN Stack HRMS Portal.

Risk ID	Risk Description	Impact Level	Probability	Mitigation Strategy
R1	MongoDB database connection failure	High	Medium	Proper configuration and backup database
R2	Login authentication errors	High	Medium	JWT testing and secure validation
R3	API integration issues	Medium	Medium	Debugging and route testing
R4	UI responsiveness problems	Medium	Low	Responsive CSS and cross-device testing
R5	Delay in project completion	High	Medium	Weekly planning and task tracking
R6	Requirement changes during development	Medium	Low	Modular coding structure
R7	Security vulnerabilities	High	Low	Password hashing, role-based access

Table4.2:Risk Assessment

4.3 Project Budget

The project budget estimates the cost involved in developing the MERN Stack HRMS Portal. Since the project was created for academic and internship purposes, the cost remained low because most technologies used were open-source and freely available.

Item	Tool/Resource	Cost
Code Editor	Visual Studio Code	Free
Frontend Framework	React.js	Free
Backend Runtime	Node.js	Free
Backend Framework	Express.js	Free
Database	MongoDB Community	Free
Version Control	GitHub	Free

Table4.3: Software and Tools Cost

Chapter 5
WORK DONE / TRAINING DETAILS

This chapter describes the actual work carried out during the internship, detailing the various modules developed, features implemented, and the key tasks performed throughout the MERN Stack HRMS Portal project.

5.1 Project Structure

The HRMS Portal was built as a monorepo with two main directories: the client (React frontend) and the server (Node.js + Express backend). The project structure was organized as follows:

- client/ – React.js frontend application with components, pages, context, and API services.
- server/ – Node.js backend with Express routes, controllers, models (Mongoose schemas), and middleware.
- server/config/ – Database connection configuration.
- server/middleware/ – JWT authentication middleware and role-based access control.
- server/models/ – Mongoose schemas for Employee, Attendance, Leave, Payroll, and Department.
- server/routes/ – Express route definitions for all API endpoints.
- server/controllers/ – Business logic for each module.

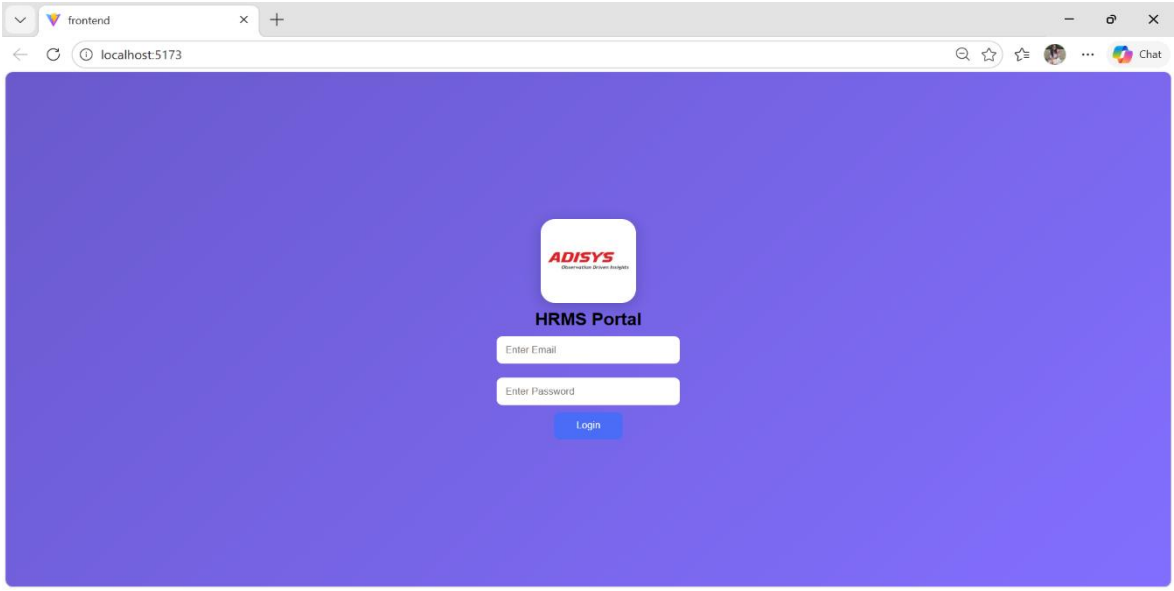

Figure 5.1– Login Page

5.2 Modules Developed

The following core HR modules were designed and developed as part of the HRMS Portal:

No.	Module Name	Features
1	Authentication Module	Login, JWT token generation, role-based access (Admin, HR, Employee)
2	Employee Management	Add, view, edit, delete employee records; assign department and role
3	Attendance Tracking	Mark attendance (Present/Absent/Leave), view attendance history, monthly reports
4	Payroll Management	Generate salary slips, calculate deductions, view payroll history
5	Department Management	Add, edit, delete departments; assign employees to departments
6	Dashboard	Role-specific dashboards with summary statistics and quick actions

Table 5.2: Developed Modules and Their Features

5.2.1 Authentication Module

The authentication module was built using JWT (JSON Web Token). When a user logs in with valid credentials, the server generates a signed JWT token that is returned to the frontend. The frontend stores this token (in localStorage or context state) and includes it in the Authorization header for all subsequent API requests. The middleware verifies the token and checks the user role before allowing access to protected routes.

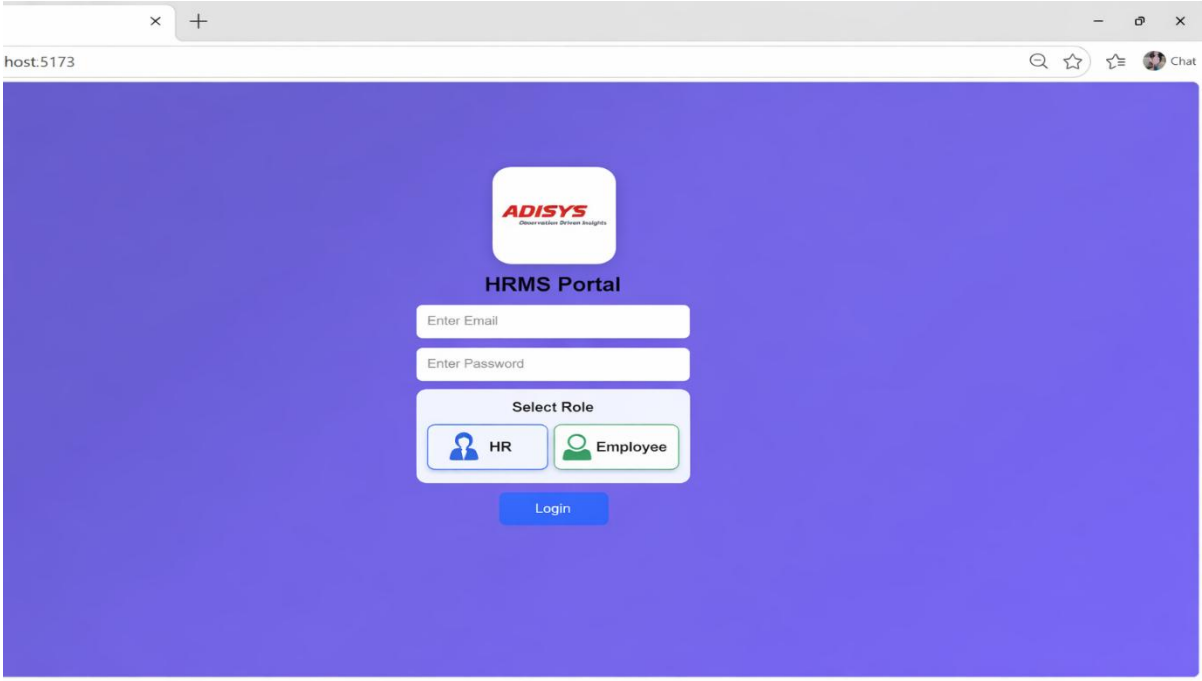

Figure 5.2.1 – Login Page with Role Selection

5.2.2 Employee Management Module

The Employee Management module allows the Admin and HR Manager to add new employees, view the full employee list, update employee details, and remove employee records. Each employee record contains personal information (name, email, phone, address), professional details (department, designation, date of joining), and login credentials. MongoDB schemas were designed using Mongoose with proper field validations.

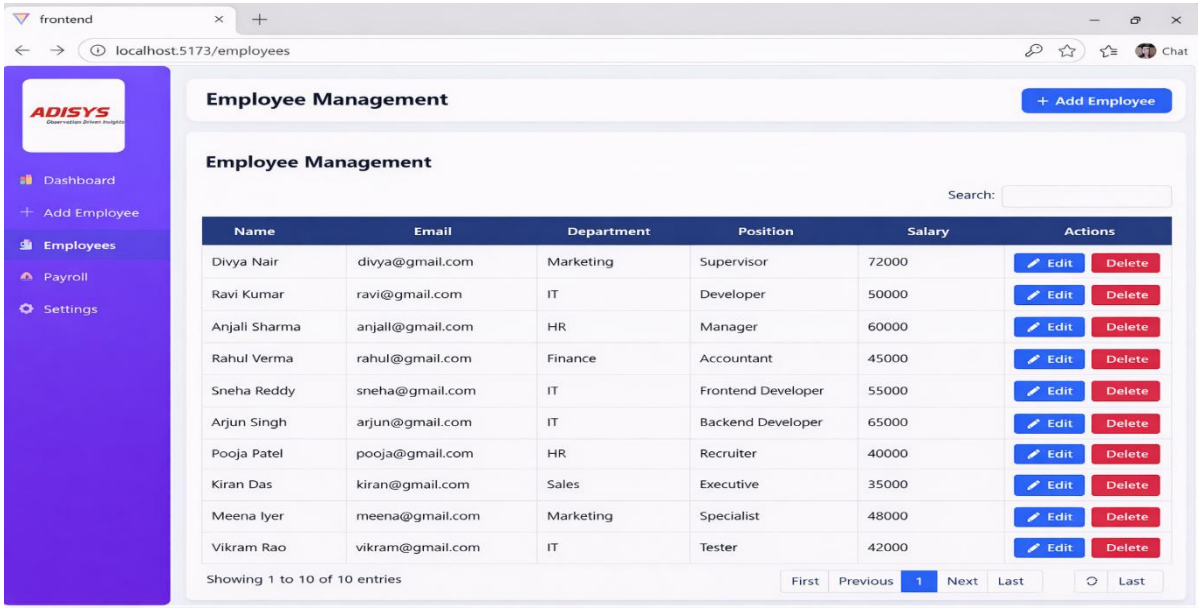

Figure 5.2.2– Employee Management Module

5.2.3 Attendance Tracking Module

The Attendance module allows HR Managers to mark daily attendance for employees, recording whether each employee is Present, Absent, or on Leave. Employees can view their own attendance records. The system displays monthly attendance summaries with counts of present, absent, and leave days. Attendance data is stored in the MongoDB Attendance collection with date, employee reference, and status fields.

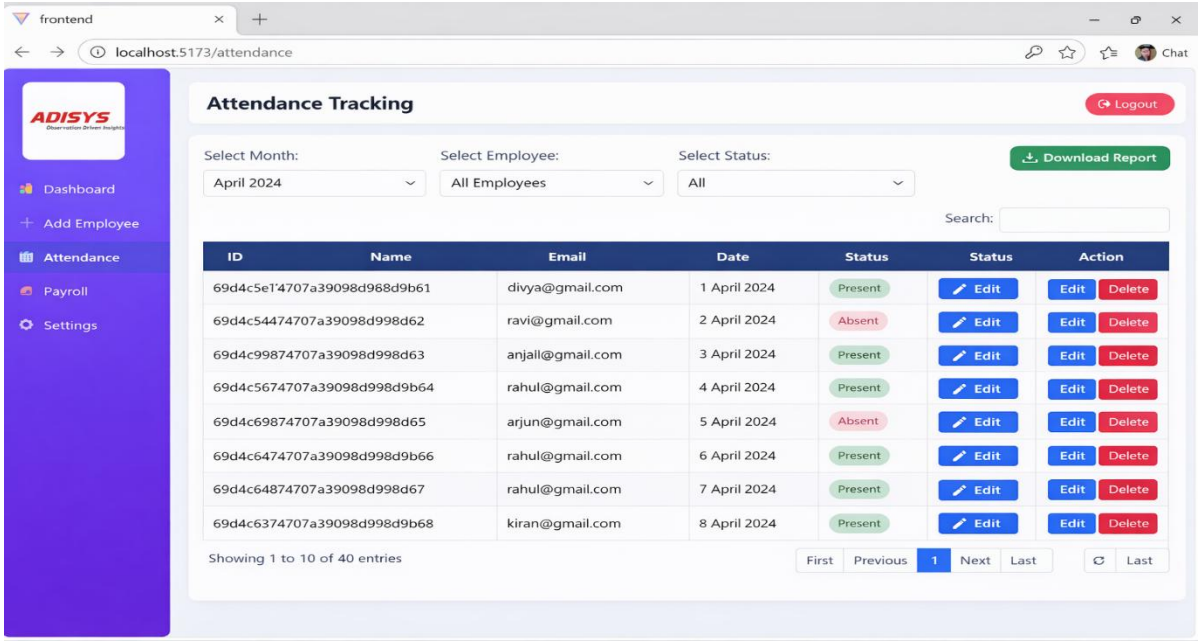

Figure 5.2.3 – Attendance Tracking Interface

5.2.4 Leave Management Module

The Leave Management module provides employees with a self-service portal to apply for leave by specifying the leave type (Casual, Sick, Paid), start and end dates, and reason. Leave requests are submitted and stored with a Pending status. HR Managers can review leave requests and approve or reject them. The system tracks the remaining leave balance for each employee and updates it accordingly after approval.

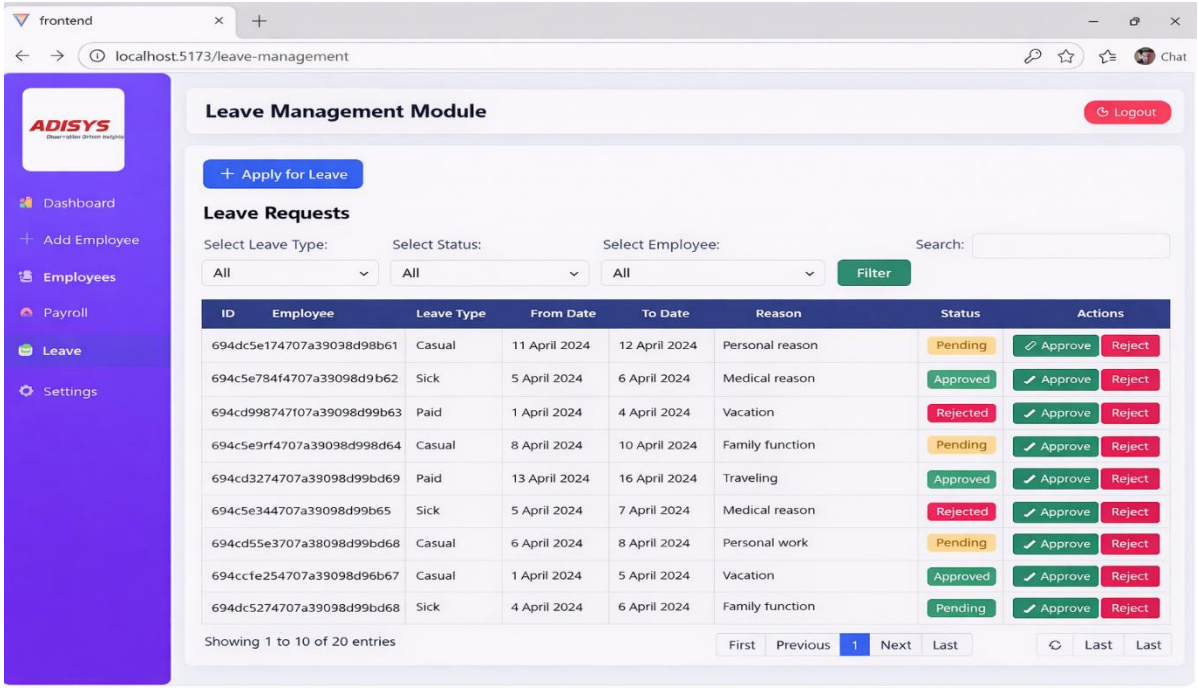

Figure 4.2.4 – Leave Management Module

5.2.5 Payroll Management Module

The Payroll module allows the Admin and HR to generate monthly salary slips for employees. The payroll calculation takes into account the base salary, attendance (deductions for absent days), allowances, and applicable deductions. Generated salary slips include a breakdown of gross pay, deductions, and net pay. Employees can view their payroll history and download salary slips from their dashboard.

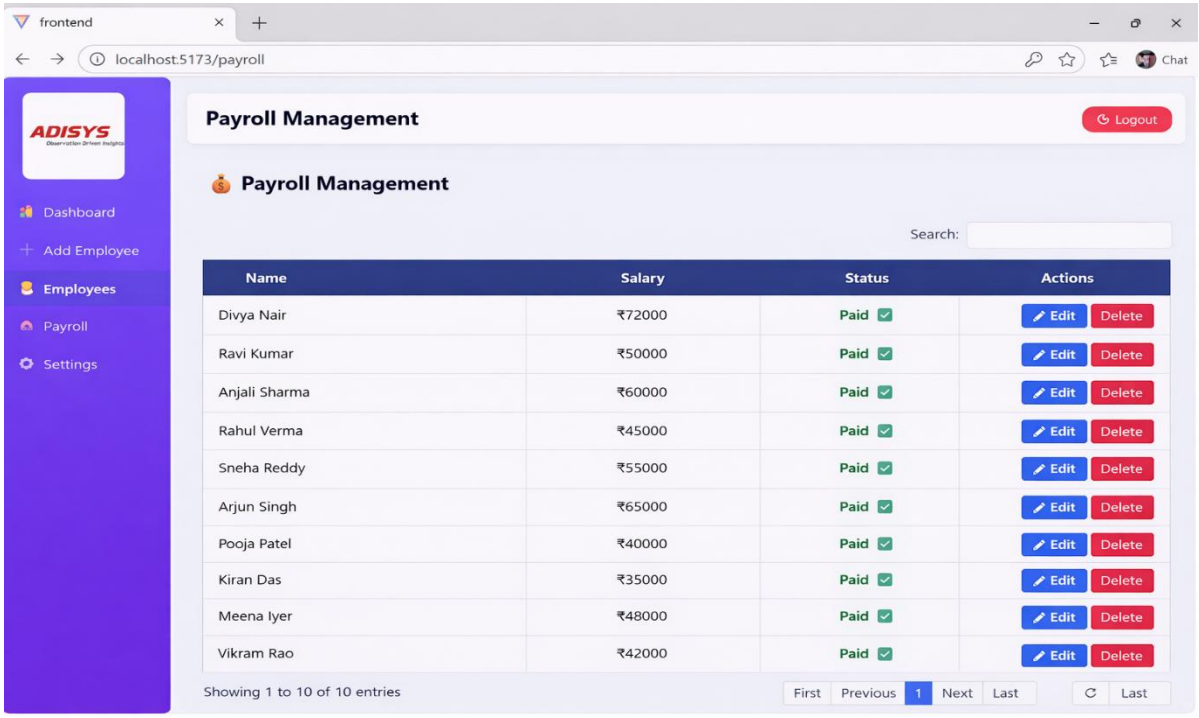

Figure 5.2.5 – Payroll Module

5.2.6 Dashboard Module

The Dashboard Module provides a role-specific landing page after successful login. The Employee Dashboard displays a welcome message with employee details, attendance summary, remaining leave balance, and salary information. It also includes scheduled meetings and recent activity such as login status, attendance marking, leave applications, and salary views. The dashboard offers quick navigation to modules such as Attendance, Leave, and Settings through the sidebar menu, providing employees with an organized and user-friendly interface.

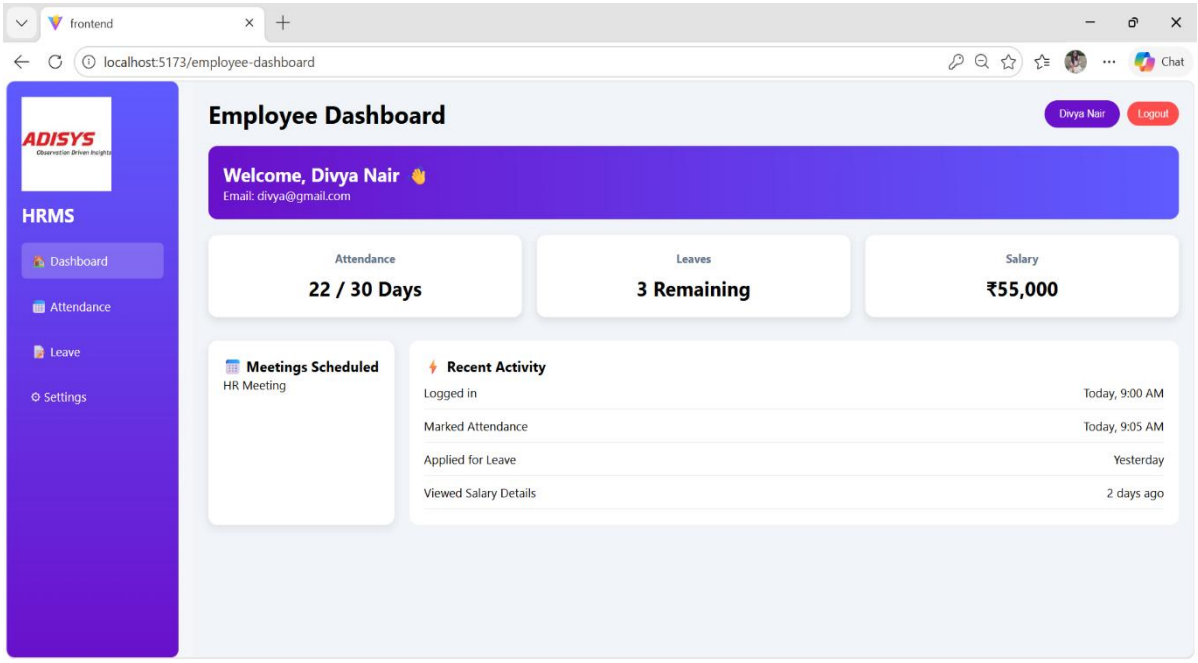

Figure 5.2.6 – Admin Dashboard

5.3 Tasks Performed

The following key tasks were performed during the development of the HRMS portal:

- Designed MongoDB database schemas for all entities including Employee, Attendance, Leave, Payroll, and Department using Mongoose.
- Built RESTful API endpoints using Express.js for all CRUD operations across all HRMS modules.
- Implemented JWT-based authentication with role-based middleware to secure API routes.
- Developed React.js components for all frontend pages including login, dashboard, employee management, attendance, leave, and payroll.
- Integrated React frontend with backend APIs using Axios for data fetching and state management.
- Implemented form validation on both frontend (React) and backend (Express) to ensure data integrity.
- Tested all API endpoints using Postman and resolved issues identified during testing.
- Applied responsive design using Tailwind CSS / Bootstrap to ensure the portal is accessible on different screen sizes.
- Used Git for version control with regular commits documenting the progress of the project.

5.4 Assignments and Practice Work

During the internship, several practice assignments were completed to strengthen understanding of the MERN stack before applying concepts to the main project. These included:

- Building a simple CRUD application using Node.js, Express.js, and MongoDB to understand RESTful API concepts.
- Creating React.js components and understanding state management using useState and useEffect hooks.
- Implementing a login system with JWT authentication as a standalone practice project.
- Practicing MongoDB queries including filtering, sorting, and aggregation pipeline operations.
- Testing API endpoints using Postman and understanding request/response formats.

5.5 Learning Approach

The internship followed a hands-on learning approach where new concepts were first studied through documentation and tutorials, and then immediately applied to the project. Node.js and Express.js documentation, React.js official documentation, and MongoDB documentation were the primary learning resources used.

Mentor guidance was sought when facing challenges related to complex API design, database modeling, or debugging issues. This blend of self-learning and guided mentorship ensured steady progress throughout the project.

5.6 Summary

This chapter described all the work performed during the internship, covering the project structure, modules developed, tasks performed, and the learning approach followed. The MERN Stack HRMS Portal was successfully developed with core HR modules including authentication, employee management, attendance tracking, leave management, payroll processing, and departmental organization.

Chapter 6

SOCIAL, LEGAL, ETHICAL, SUSTAINABILITY AND SAFETY ASPECTS

Building a web-based HRMS portal has significant implications across social, legal, ethical, sustainability, and safety dimensions. This chapter explores how the MERN Stack HRMS Portal addresses each of these important aspects.

6.1 Social Aspects

6.1.1 Improving Workplace Efficiency

The HRMS portal directly benefits organizations and their employees by automating repetitive HR tasks. Employees can apply for leave, check attendance, and view salary information independently without depending on HR staff for every query. This self-service model saves time, reduces miscommunication, and improves overall workplace efficiency.

6.1.2 Promoting Digital Inclusion

By providing a web-based, device-agnostic HRMS solution accessible from any browser, the system promotes digital inclusion across the workforce. Employees at all levels, from executives to field workers, can access HR services using the same interface, regardless of their location or device.

6.1.3 Supporting Employee Well-Being

Efficient leave management, transparent payroll processing, and accurate attendance tracking contribute to employee satisfaction and well-being. When employees have clear visibility into their HR records and receive timely payroll, it reduces workplace stress and promotes a healthy work environment.

6.2 Legal Aspects

6.2.1 Data Privacy and Confidentiality

The HRMS portal handles sensitive personal and financial information of employees, including salary details, personal addresses, contact numbers, and attendance records. The application ensures that this data is protected through role-based access control, meaning that employees can only view their own data, while HR and Admin have appropriate access levels.

6.2.2 Compliance with Labour Laws

The leave management and payroll modules are designed to align with organizational HR policies and general labour law guidelines, such as mandatory leave entitlements and salary payment norms. Organizations implementing the HRMS can configure these settings to remain compliant with applicable regulations.

6.2.3 Intellectual Property

All code and design created during this internship project is original work developed as part of the academic internship. Third-party libraries and frameworks such as React.js, Express.js, MongoDB, and JWT are open-source tools used under their respective open-source licenses (MIT, Apache, etc.), ensuring legal compliance in technology usage.

6.3 Ethical Aspects

6.3.1 Fair Access and Transparency

The HRMS portal promotes fairness by providing all employees equal access to their personal HR records. The payroll and attendance systems are transparent, ensuring that calculations are based on accurate data and clearly displayed to employees, reducing potential disputes.

6.3.2 Responsible Data Handling

During the development of the HRMS portal, ethical responsibility in data handling was prioritized. Test data used during development was anonymized and did not include real personal information. The system is designed to store only the minimum necessary data required for HR operations.

6.3.3 Professional Integrity

Throughout the internship, professional integrity was maintained by writing clean, documented, and honest code. No proprietary code or confidential algorithms were used without proper authorization. All sources of reference, third-party libraries, and tutorials used during development are acknowledged appropriately.

6.4 Sustainability Aspects

6.4.1 Reducing Paper Usage

The digitization of HR processes through the HRMS portal eliminates the need for paper-based record keeping, attendance registers, salary slips, and leave forms. This contributes directly to environmental sustainability by reducing paper consumption in organizations.

6.4.2 Scalable and Maintainable Architecture

The MERN stack architecture used for this project is highly scalable, allowing organizations to expand the system as their workforce grows without requiring a complete rebuild. The modular design ensures long-term maintainability, reducing the need for costly system overhauls.

6.4.3 Supporting Sustainable Workforce Management

Efficient HR management enabled by the HRMS portal helps organizations optimize their workforce utilization, reduce administrative overhead, and make data-driven decisions, contributing to the long-term economic sustainability of the organization.

6.5 Safety Aspects

6.5.1 Application Security

Security was a primary concern throughout the development of the HRMS portal. JWT-based authentication ensures that only verified users can access the system. Role-based access control ensures that users can only access resources appropriate to their role, preventing unauthorized data access.

6.5.2 Password Security

User passwords are stored in hashed form using bcrypt, a widely recognized password hashing algorithm. This ensures that even in the unlikely event of a database breach, actual passwords remain secure and cannot be reverse-engineered.

6.5.3 Input Validation and Error Handling

All API endpoints are protected with input validation to prevent common web vulnerabilities such as SQL injection (not applicable to MongoDB, but analogous NoSQL injection), XSS, and improper data formats. Proper error handling ensures that sensitive system information is not exposed to users through error messages.

6.6 Summary

This chapter highlighted the social, legal, ethical, sustainability, and safety considerations addressed in the MERN Stack HRMS Portal project. The system promotes workplace efficiency, data transparency, and security while adhering to ethical development practices and contributing to organizational and environmental sustainability.

Chapter 7
CHALLENGES FACED

During the development of the MERN Stack HRMS Portal, several technical and non-technical challenges were encountered. These challenges provided valuable learning opportunities and helped in developing problem-solving skills. The following are the key challenges faced and how they were addressed:

- **JWT Token Management:** Managing JWT token expiration and refreshing sessions was initially challenging. This was resolved by implementing proper token expiry checks on the backend and redirecting users to the login page upon token expiration on the frontend.
- **Role-Based Access Control Implementation:** Designing and implementing role-based middleware that correctly restricts access to different API routes based on user roles (Admin, HR Manager, Employee) required careful planning. The challenge was resolved by designing a middleware function that verifies the JWT token and checks the user's role field before granting access to protected routes.
- **Mongoose Schema Design:** Designing efficient MongoDB schemas with proper relationships (using object references) between collections such as Employee, Attendance, Leave, and Department required careful consideration. The challenge of data referencing vs. embedding was addressed through analysis of data access patterns for each module.
- **CORS Configuration:** Cross-Origin Resource Sharing (CORS) errors between the React frontend (running on port 3000) and the Express backend (running on port 5000) were encountered during development. This was resolved by properly configuring the CORS middleware in Express.
- **State Management in React:** Managing complex application state across multiple components (especially user authentication state and HR data) was challenging. This was addressed using React Context API for global state management, avoiding excessive prop drilling.

- **Payroll Calculation Logic:** Implementing the payroll calculation logic that accounts for attendance, allowances, and deductions required careful mathematical formulation. The logic was carefully designed, tested with multiple test cases, and validated against expected results.

- **Responsive Design:** Making all HR module pages responsive for different screen sizes required additional effort in CSS styling. Tailwind CSS utility classes were used effectively to achieve responsive layouts.

All the above challenges were gradually overcome through systematic debugging, consultation of documentation, online community resources (MDN, Stack Overflow, official docs), and guidance from mentors.

Chapter 8
LEARNING OUTCOMES

The internship on MERN Stack HRMS Portal Development has provided comprehensive learning across multiple dimensions of full-stack web development and professional software engineering. The following are the key learning outcomes achieved through this internship:

- **Full-Stack Development Expertise:** Gained practical knowledge of the complete MERN stack (MongoDB, Express.js, React.js, Node.js) by building a production-grade web application from scratch.

- **RESTful API Design:** Learned the principles of REST architecture and developed RESTful APIs with proper HTTP methods (GET, POST, PUT, DELETE), status codes, and JSON response formats.

- **Database Design and Management:** Developed skills in designing NoSQL database schemas using Mongoose, including understanding of data modeling strategies such as referencing and embedding in MongoDB.

- **Authentication and Security:** Gained understanding of JWT-based authentication, password hashing using bcrypt, and role-based access control (RBAC) to secure web applications.

- **React.js Component Development:** Learned to build reusable and responsive UI components using React.js, manage component state with hooks (useState, useEffect), and handle routing with React Router.

- **API Integration:** Developed skills in integrating frontend and backend applications using Axios for HTTP requests, handling API responses, and managing loading and error states.

- **Version Control with Git:** Gained hands-on experience in using Git for version control, making regular commits, working with branches, and managing a GitHub repository for the project.

- **API Testing:** Learned to test and validate RESTful APIs using Postman, including creating test collections, setting authorization headers, and verifying API responses.

- **Problem-Solving Skills:** Developed the ability to independently debug and resolve technical issues through systematic analysis and use of available resources.

- **Professional Development:** Improved technical documentation skills, time management, and the ability to work on a structured software development project within a defined timeline.

Chapter 9
CONCLUSION

This chapter summarizes the overall experience, achievements, and insights gained during the MERN Stack HRMS Portal Development internship.

9.1 Summary of Work Undertaken

The internship involved the end-to-end design and development of a full-stack Human Resource Management System (HRMS) Portal using the MERN stack. The project was divided into systematic phases including requirement analysis, system design, backend API development, frontend UI development, integration, and testing.

The developed HRMS portal includes a complete set of HR management modules: Authentication with JWT and RBAC, Employee Management, Attendance Tracking, Leave Management, Payroll Processing, Department Management, and role-specific Dashboards. The system was built with a three-tier MERN architecture ensuring scalability, security, and maintainability.

9.2 Achievement of Objectives

All the primary objectives of the internship project were successfully achieved. A fully functional HRMS web application was built using the MERN stack with role-based access control for Admin, HR Manager, and Employee roles. All five core HR modules were developed and integrated into a working system. JWT authentication, bcrypt password hashing, and input validation were implemented to ensure application security. The frontend was built with a responsive design accessible from different devices and browsers.

9.3 Key Learnings

The internship provided invaluable practical experience in full-stack web development using modern technologies. Key learnings include deep understanding of the MERN architecture, RESTful API design patterns, JWT-based security, MongoDB schema design, React.js component development, and professional software development practices including version control and API testing.

Beyond technical skills, the internship also developed essential professional skills including structured project management, technical documentation, problem-solving under constraints, and the ability to independently learn and apply new technologies.

9.4 Limitations

While the HRMS portal covers all core HR modules, there are some limitations in the current version. Advanced HR features such as performance management, recruitment tracking, and training management were not included in this version due to time constraints. The payroll module uses simplified salary calculation logic and may require customization for organizations with complex payroll structures.

The application currently uses basic MongoDB queries without advanced aggregation pipelines, which may need optimization for large-scale data. Real-time notifications for leave approvals and payroll generation were not implemented in the current version.

9.5 Recommendations

Students planning to undertake internships in full-stack web development should build foundational knowledge in JavaScript, HTML, and CSS before starting. Familiarity with basic database concepts and API fundamentals will also greatly accelerate the learning process during the internship.

For future development of the HRMS portal, the following enhancements are recommended:

- Integration of real-time notifications using Socket.io for leave approval alerts.
- Addition of performance management and appraisal tracking modules.
- Implementation of reporting and analytics dashboards with charts using libraries such as Chart.js or Recharts.
- Cloud deployment on platforms such as AWS, Heroku, or Vercel with MongoDB Atlass.
- Mobile application development using React Native for on-the-go HR access.

9.6 Concluding Remarks

The MERN Stack HRMS Portal Development internship has been an enriching and rewarding experience that provided deep practical exposure to modern full-stack web development. Building a complete, functional HRMS system from scratch using MongoDB, Express.js, React.js, and Node.js has significantly enhanced technical competence, project management skills, and professional readiness for a career in software development.

The knowledge and experience gained during this internship will serve as a strong foundation for future professional endeavors in web development and software engineering.

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[7] Postman Documentation – For API testing and validation.
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[8] Tailwind CSS Documentation – For responsive UI design.
Available: https://tailwindcss.com/docs

[9] Git Documentation – For version control best practices.
Available: https://git-scm.com/doc

APPENDIX

Appendix A – API Endpoints Reference

Method	Endpoint	Module	Description
POST	/api/auth/login	Auth	User login and JWT generation
GET	/api/employees	Employee	Get all employees (Admin/HR)
POST	/api/employees	Employee	Add new employee
PUT	/api/employees/:id	Employee	Update employee details
DELETE	/api/employees/:id	Employee	Delete employee record
GET	/api/attendance	Attendance	Get attendance records
POST	/api/attendance	Attendance	Mark attendance
GET	/api/leaves	Leave	Get all leave requests
POST	/api/leaves	Leave	Apply for leave
PUT	/api/leaves/:id	Leave	Approve/Reject leave
GET	/api/payroll	Payroll	Get payroll records
POST	/api/payroll	Payroll	Generate payroll
GET	/api/departments	Department	Get all departments
POST	/api/departments	Department	Add new department

Appendix B – MongoDB Schema Design

The following collections were designed in MongoDB for the HRMS Portal:

- users – Stores login credentials (name, email, password hash, role)
- employees – Stores employee personal and professional details
- departments – Stores department names and descriptions
- attendances – Stores daily attendance records (employeeId, date, status)
- leaves – Stores leave requests (employeeId, type, dates, status, reason)
- payrolls – Stores payroll records (employeeId, month, basicSalary, deductions, netPay)